

XRPL Validator — Daily Engineering Report • 2026-03-21

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **36 commits** (xrpl-sdk 10, xrpl-validator 26)

Engineering changes

- **fix: correct repository URL in all Cargo.toml files** (4417b88 , xrpl-sdk) Update repository field from placeholder xrpl-rust org to actual ilubxrp589 account. Fixes docs.rs links, crates.io metadata, and cargo tree output.
- **fix(core): update thread_rng() to rng() for rand 0.9 compatibility** (fb7a327 , xrpl-sdk) rand 0.9 renamed thread_rng() to rng().
- **feat: initial scaffold — xrpl-ledger and xrpl-node crates** (4299395 , xrpl-validator) Two-crate workspace for building an XRPL validator node in Rust: - xrpl-ledger: SHAMap, NodeStore trait, ledger header/objects/state (module stubs + error types, depends on xrpl-core via git) - xrpl-node: peer protocol, consensus, mempool, RPC server (module stubs + config + error types, protobuf codegen from rippled's xrpl.proto, tracing framework)
- **feat(node): node identity, message types, and wire codec** (07c32fd , xrpl-validator)
- peer/identity.rs: NodeIdentity with Ed25519 keypair, self-signed TLS cert (pubkey in CN), rustls server/client configs, AcceptAnyCert verifier for self-signed peer network (4 tests) - peer/message.rs: PeerMessage enum wrapping all 21 protobuf message types with encode/decode dispatch by type code - peer/codec.rs: MessageCodec implementing tokio-util Decoder/Encoder with 4-byte length prefix, 2-byte type code, zlib ...
- **feat(node): handshake, connection, peer lifecycle** (d92281f , xrpl-validator)
- peer/handshake.rs: outbound XRPL/2.0 handshake over TLS — HTTP upgrade with Public-Key, Session-Signature (TLS exporter keying material), Network-ID, Crawl headers. Supports mainnet/testnet/devnet. - peer/connection.rs: async read/write loops via tokio select!, auto ping/pong handling, overlay message forwarding, graceful shutdown via watch channel. PeerHandle for sending messages to peers. PeerInfo tracks pubkey, ...
- **feat(node): complete M1 peer protocol — live testnet handshake working** (ebbb373 , xrpl-validator) Major milestone: successfully handshake with XRPL testnet and decode peer protocol messages.
- **fix(node): M1 audit fixes — remaining bytes, docs, codec safety** (140d13d , xrpl-validator) Fixes from peer protocol audit: 1. CRITICAL: Pass remaining_bytes from handshake through to connection's codec buffer. The HTTP header read may consume bytes belonging to the binary protocol — these must be prepended to the Framed decoder's buffer.
- **fix(codec): correct frame length interpretation — messages now decode** (12f9941 , xrpl-validator) The 4-byte length field in XRPL peer protocol frames is the protobuf payload size ONLY — it does NOT include the 2-byte

type code.

- **feat(node): live XRPL peer message viewer on port 3777** (e9613c4 , xrpl-validator) Web dashboard streaming real-time decoded XRPL peer protocol messages from the testnet via Server-Sent Events.
- **feat(viewer): add built-with credit line** (f1cc9dc , xrpl-validator)
- **fix(viewer): no-scroll page layout with contained feed box** (2f3633d , xrpl-validator) Page no longer scrolls — feed is in a bordered box that scrolls internally. Header and built-with text stay fixed.
- **feat(viewer): add (TESTNET) label to header** (b7b30c0 , xrpl-validator)
- **feat(ledger): SHAMap implementation — hash prefixes, nodes, trie** (04f795c , xrpl-validator) M2 progress: core SHAMap data structure for XRPL ledger state.
- **feat(ledger): complete M2 — SHAMap, NodeStore, ledger state (audited)** (1b424d4 , xrpl-validator) Full ledger storage implementation, audited twice with all findings fixed.
- **feat(viewer): Ripple-branded dashboard with live visualizations** (8d2cc9b , xrpl-validator) Complete UI overhaul with XRP/Ripple brand colors (#0085FF primary, dark navy backgrounds) and Orbitron + JetBrains Mono typography.
- **feat(viewer): peer connection list display** (0dbac6e , xrpl-validator) Backend sends peer endpoint data from TMEndpoints messages via SSE. Frontend displays peers as chips in a bar between the type distribution and stats row. Shows endpoint address and hop count. Updates each time the connected peer sends its peer list (~every 60s).
- **feat(viewer): live validator consensus grid** (e3f904a , xrpl-validator) Each validator is now tracked individually during consensus: - Proposals: extract node_pub_key from TMProposeSet, dot turns blue - Validations: extract SigningPubKey from serialized validation blob by scanning for field code 0x73 0x21, dot turns yellow - Consensus reached (ACCEPTED): all active dots flash green
- **feat(viewer): dedicated consensus monitor page at /consensus** (a4ffd3c , xrpl-validator) Full-screen validator agreement visualization: - Big ledger counter (Orbitron 48px, blue flash on close) - Phase indicator: PROPOSING → CLOSING → VALIDATING → CONSENSUS - Validator grid: 16x16px nodes that light up as they participate - Idle (dark) → Proposed (blue) → Validated (yellow) → Agreed (green) - Tooltip on hover shows validator key prefix - Stats: proposals, validations, unique validators seen, close ...
- **feat(viewer): nebula background, physics orbs, rainbow consensus** (853e04e , xrpl-validator) Complete consensus page redesign: - Animated nebula background (5 drifting gradient blobs, 80px blur) - Force-directed validator orbs with orbital spin, gravity, repulsion - State effects: proposed (blue burst), validated (yellow shockwave), agreed (green explosion + outward blast + gravity pull-back) - Rainbow particle explosion on consensus: 80 HSL-cycling particles from center + 12 per agreed node, with glow ...
- **feat(node): complete M3 transaction engine with live sig verification** (604c1c1 , xrpl-validator) Full transaction validation pipeline with codec integration (audited): mempool/validate.rs: - Decode tx blob via xrpl-core binary codec - Extract Account, Fee, Sequence, LastLedgerSequence from decoded JSON - Signature verification: re-encode for signing, compute signing hash - Ed25519: verify against raw STX\0||bytes (Ed25519 does internal hash) - Secp256k1: verify against SHA512Half(STX\0||bytes) - Fee sanity ...
- **feat(node): complete M4 consensus engine (audited)** (bf78b0a , xrpl-validator) Full RPCA consensus implementation: - unl.rs: Trusted validator list, manifest key rotation - state.rs: Open/Establish/Accept state machine with stability check - threshold.rs: Dynamic convergence 50% → 60% → 70% → 80% → 95% - close_time.rs: Resolution rounding, mode voting - validation.rs: Per-ledger validation tracking, >=80% fully-validated - proposal.rs: Transaction set hash (single source, re-exported) - ...

- **feat: complete all 40 cards + metrics page + full audit fixes** ([ab2ad99](#) , [xrpl-validator](#)) Final milestone — all 40 taskboard cards implemented and audited.
- **feat: switch to XRPL mainnet + domain verification** ([48544b2](#) , [xrpl-validator](#)) val-041: Switched from testnet to mainnet - Peer: s1.ripple.com:51235 (was s.altnet.rippletest.net) - Network-ID: 0 (mainnet, was 1 testnet) - All 3 pages updated: MAINNET label
- **feat: sled storage, PM2 service, 3D consensus, GitHub link** ([eec04a2](#) , [xrpl-validator](#)) val-043: Sled persistent NodeStore behind 'sled-backend' feature val-054: PM2 'xrpl-validator' service on release binary
- **feat(node): fast parallel ledger sync — 1500+ objects/sec from mainnet** ([c93fef7](#) , [xrpl-validator](#)) val-044: sync_ledger binary downloads XRPL mainnet state objects.
- **feat(viewer): tx/s rate on feed + metrics pages** ([cb4a1ab](#) , [xrpl-validator](#)) Feed page: tx/s in stats row, updated every second Metrics page: Transactions/Second card with purple sparkline graph
- **feat(viewer): live ledger sync progress bar on feed + metrics** ([3dc97a2](#) , [xrpl-validator](#)) /api/sync-status endpoint reads sync file from NVMe drive, returns object count, estimated percentage, file size, and active status.
- **feat(viewer): multi-peer mainnet connections** ([59d512f](#) , [xrpl-validator](#)) Connect to s1.ripple.com AND s2.ripple.com simultaneously. Each peer runs its own connection loop with auto-reconnect. Both feed into the same SSE stream — double the network visibility.

Docs and operations

- **docs: replace "production-grade" with "community-built"** ([b2d8b35](#) , [xrpl-sdk](#))
- **ci: add Dependabot for cargo and GitHub Actions updates** ([3710bc3](#) , [xrpl-sdk](#))
- **ci: bump actions/checkout from 4 to 6** ([14b6e1b](#) , [xrpl-sdk](#))
- **deps: bump tokio-tungstenite from 0.26.2 to 0.29.0** ([753529d](#) , [xrpl-sdk](#))
- **deps: bump rand from 0.8.5 to 0.9.2** ([1ba8be6](#) , [xrpl-sdk](#))
- **Merge pull request #1 from ilubxrp589/dependabot/github_actions/actions/checkout-6** ([a9f1a34](#) , [xrpl-sdk](#))
- **Merge pull request #2 from ilubxrp589/dependabot/cargo/tokio-tungstenite-0.29.0** ([6117e6a](#) , [xrpl-sdk](#))
- **Merge pull request #3 from ilubxrp589/dependabot/cargo/rand-0.9.2** ([360662e](#) , [xrpl-sdk](#))

Appendix — commits

Time	Repo	Commit	Subject
11:51	xrpl-sdk	4417b88	fix: correct repository URL in all Cargo.toml files
11:53	xrpl-sdk	b2d8b35	docs: replace "production-grade" with "community-built"
12:05	xrpl-sdk	3710bc3	ci: add Dependabot for cargo and GitHub Actions updates
12:06	xrpl-sdk	14b6e1b	ci: bump actions/checkout from 4 to 6
12:06	xrpl-sdk	753529d	deps: bump tokio-tungstenite from 0.26.2 to 0.29.0
12:06	xrpl-sdk	1ba8be6	deps: bump rand from 0.8.5 to 0.9.2
15:33	xrpl-sdk	fb7a327	fix(core): update thread_rng() to rng() for rand 0.9 compatibility
15:35	xrpl-sdk	a9f1a34	Merge pull request #1 from ilubxrp589/dependabot/github_actions/actions/checkout-6
15:35	xrpl-sdk	6117e6a	Merge pull request #2 from ilubxrp589/dependabot/cargo/tokio-tungstenite-0.29.0
15:35	xrpl-sdk	360662e	Merge pull request #3 from ilubxrp589/dependabot/cargo/rand-0.9.2

Time	Repo	Commit	Subject
16:26	xrpl-validator	4299395	feat: initial scaffold — xrpl-ledger and xrpl-node crates
16:32	xrpl-validator	07c32fd	feat(node): node identity, message types, and wire codec
16:39	xrpl-validator	d92281f	feat(node): handshake, connection, peer lifecycle
16:59	xrpl-validator	ebbb373	feat(node): complete M1 peer protocol — live testnet handshake working
17:13	xrpl-validator	140d13d	fix(node): M1 audit fixes — remaining bytes, docs, codec safety
17:23	xrpl-validator	12f9941	fix(codec): correct frame length interpretation — messages now decode
17:28	xrpl-validator	e9613c4	feat(node): live XRPL peer message viewer on port 3777
17:31	xrpl-validator	f1cc9dc	feat(viewer): add built-with credit line
17:33	xrpl-validator	2f3633d	fix(viewer): no-scroll page layout with contained feed box
17:43	xrpl-validator	b7b30c0	feat(viewer): add (TESTNET) label to header
18:48	xrpl-validator	04f795c	feat(ledger): SHAMap implementation — hash prefixes, nodes, trie
19:10	xrpl-validator	1b424d4	feat(ledger): complete M2 — SHAMap, NodeStore, ledger state (audited)
19:16	xrpl-validator	8d2cc9b	feat(viewer): Ripple-branded dashboard with live visualizations
19:31	xrpl-validator	0dbac6e	feat(viewer): peer connection list display
19:48	xrpl-validator	e3f904a	feat(viewer): live validator consensus grid
19:52	xrpl-validator	a4ffd3c	feat(viewer): dedicated consensus monitor page at /consensus
20:41	xrpl-validator	853e04e	feat(viewer): nebula background, physics orbs, rainbow consensus
21:17	xrpl-validator	604c1c1	feat(node): complete M3 transaction engine with live sig verification
21:31	xrpl-validator	bf78b0a	feat(node): complete M4 consensus engine (audited)
22:00	xrpl-validator	ab2ad99	feat: complete all 40 cards + metrics page + full audit fixes
22:14	xrpl-validator	48544b2	feat: switch to XRPL mainnet + domain verification
22:47	xrpl-validator	eec04a2	feat: sled storage, PM2 service, 3D consensus, GitHub link
23:18	xrpl-validator	c93fef7	feat(node): fast parallel ledger sync — 1500+ objects/sec from mainnet
23:41	xrpl-validator	cb4a1ab	feat(viewer): tx/s rate on feed + metrics pages
23:48	xrpl-validator	3dc97a2	feat(viewer): live ledger sync progress bar on feed + metrics
23:59	xrpl-validator	59d512f	feat(viewer): multi-peer mainnet connections